SectraTEK

AI FOR CYBERSECURITY AND DIGITAL FORENSICS · LAB 06

Answers You Can Cite



Retrieval-Augmented Generation: ask plain-English questions over your forensic reports and get an answer grounded in them, with citations.

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YOUR INSTRUCTOR

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IN ONE LINE

I study how phones get spied on and tampered with, and build AI tools that help examiners find the evidence.

Mobile threat research
Spyware, stalkerware and device tampering.

AI for investigators
Machine learning that speeds up the search.

Teaching
AI for Cybersecurity and Digital Forensics.

TODAY'S STORY

From a pile of reports to a cited answer

1

Background

An examiner's questions span many past cases.

2

Before AI

Keyword search finds words, not meaning.

3

The problem

Ask an LLM directly and it invents your cases.

4

The gap

General knowledge is not your evidence.

5

How RAG helps

Retrieve the right passages, then answer from them.

6

The method

Chunk, embed, retrieve, generate.

7

The case

Six reports, one grounded, cited answer.

8

Your lab

Build the pipeline by hand in the browser.

Practical runs in Google Colab with a free hosted model. Nothing is installed on your computer.

BACKGROUND

The examiner's growing archive

A working examiner accumulates hundreds of case reports. The useful questions cut across all of them, and the answers are buried in the text.

Many cases

Reports pile up: mobile, malware, memory, disk, network.

Cross-case questions

“Which cases involved ransomware?” spans the whole archive.

Slow to answer

Finding the answer means remembering, then re-reading, report by report.

BEFORE AI

Keyword search finds words, not meaning

YOU SEARCH FOR		YOU MISS
“credentials”	⇄	“LSASS dump”, “password theft”, “Mimikatz”
“ransomware”	⇄	“files encrypted”, “ransom note”, “.lxk extension”
“exfiltration”	⇄	“copied to Telegram”, “see attached export”

Keyword search matches letters. Two reports can describe the same thing in completely different words.

THE PROBLEM

Ask the model directly, and it invents your case

ASK AN LLM ABOUT CASE DF-2026-025

"Case DF-2026-025 involved a phishing email that deployed Emotet, exfiltrating 2.4 GB over Tor on March 9th."

EVERY SPECIFIC IS INVENTED

It never saw your report

The model was trained on the public web, not your case files.

So it fills the blanks

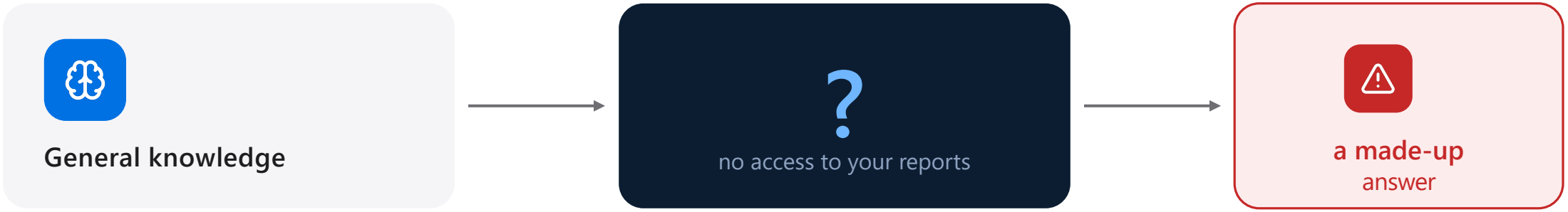
Asked about a case it cannot know, it produces a plausible story.

Confident and wrong

Emotet, 2.4 GB, Tor, March 9th: none of it is in the real report.

THE GAP

The model has knowledge, not your evidence

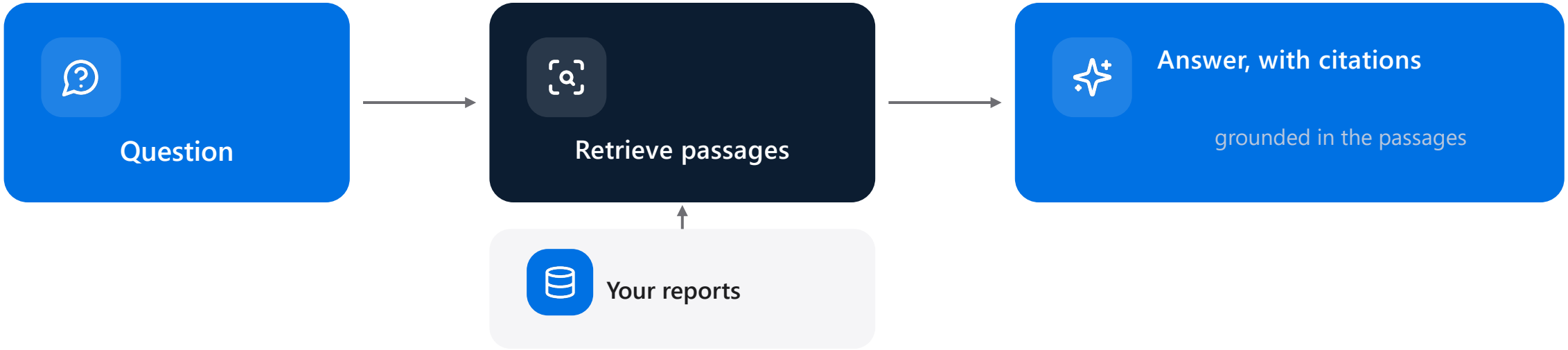


WE NEED TO HAND IT THE RIGHT EVIDENCE, AT QUESTION TIME

The fix is not a bigger model. It is to find the few report passages that actually answer the question, and give the model only those, with an instruction to answer from them and cite them. That is Retrieval-Augmented Generation.

THE ANSWER

Retrieve, then generate



THE MODEL ONLY SEES WHAT YOU RETRIEVE

It answers from the passages you found, not from memory. That is what keeps it honest.

AND IT CITES

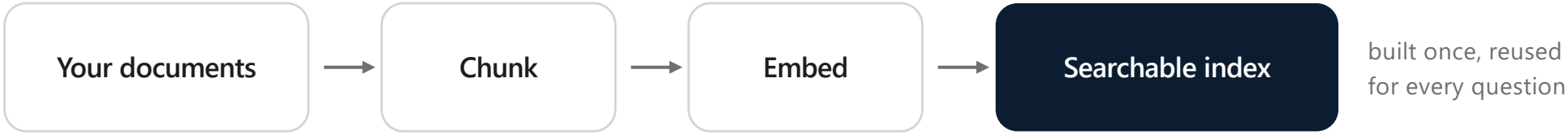
Each claim names the report it came from, so you can check it against the source.

THE SHAPE OF RAG

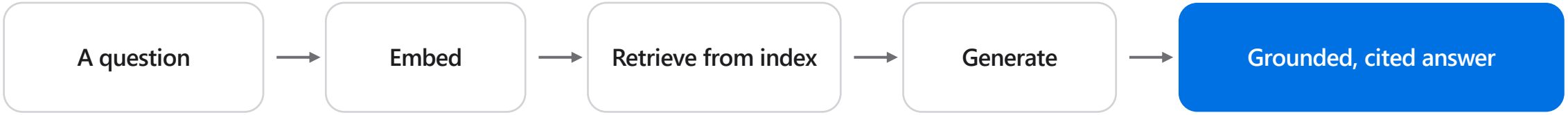
Set up once, then ask it anything

RAG has two phases. You build a searchable index from your documents one time, then every question reuses that index to find and answer.

ONE-TIME SETUP



EACH QUESTION (reuses the index above)



IN GOES YOUR EVIDENCE AND A QUESTION; OUT COMES A CITED ANSWER

The model answers only from the passages retrieved from your index, and names the document behind each claim.

HOW IT WORKS

Four steps, built by hand



Chunk

Cut each report into short passages, each tagged with its source.



Embed

Turn each passage into numbers that capture its meaning.



Retrieve

Embed the question, find the closest passages by meaning.



Generate

Answer from those passages only, and cite the report.

YOU BUILD EACH STEP YOURSELF

No framework hides the pipeline. In the notebook you chunk, embed, and search with a few lines of plain Python, so you can see exactly how a passage is found and how the answer is grounded.

THE ONE NEW IDEA

An embedding is meaning as numbers

An embedding turns a piece of text into a list of numbers (a vector). Text with similar meaning gets similar numbers, so “close numbers” means “close meaning.”

“the attacker dumped credentials from LSASS”	close
“passwords were stolen from memory”	close
“the phone was left on a supply table”	far

WHY IT MATTERS

The first two lines never share a keyword, but they mean the same thing, so their embeddings sit close together. That is how retrieval finds the right passage even in different words.

A small, free model does the embedding, right in your notebook session.

THE PRACTICAL

Six reports, one question box

YOUR KNOWLEDGE BASE

- Mobile data exfiltration
- Windows ransomware
- Memory credential theft
- iOS stalkerware
- Disk deleted-file recovery
- Network intrusion

WHAT YOU DO

Build the index

Chunk and embed all six reports into a searchable set of passages.

Ask across cases

“Which cases showed anti-forensic behaviour?” finds the right reports.

Get a cited answer

The model answers from the retrieved passages and names each source.

SEE THE DIFFERENCE

The same question, two ways

Model alone

no reports

"DF-2026-025 involved Emotet exfiltrating 2.4 GB over Tor."

Fluent, specific, and invented. No source to check.

With RAG

retrieved + cited

"DF-2026-025 used a Mimikatz-like tool to dump credentials from LSASS, kept alive by a scheduled task."

Grounded in the report, and it cites [DF-2026-025].

PRACTICAL

Your turn, in the browser

1	**Part 0: Set up**	0.1 install, 0.2 connect to Groq with your free key, 0.3 save the six reports.
2	**Part A: Build it by hand**	A.1 split into passages, A.2 embed, A.3 retrieve, A.4 answer with citations.
3	**Part B: Why retrieval matters**	B.1 ask the model alone, B.2 ask the same question with retrieval.
4	**Part C: Your own questions**	C.1 ask across all six reports, and read the sources it retrieved.
5	**Write up and submit**	Confirm one citation, find a question it cannot answer, save the notebook.

Open it in Colab from the badge. Every step explains its Purpose, Input, How it works, and Output.

SectraTEK

Let the reports speak for themselves.

RAG turns a pile of case files into a question box that answers only from the evidence, and shows its sources.

